Reproduction Report

The Narrow-Channel Effect in MOSFET's with Semi-Recessed Oxide Structures

Li, Hong, Cheng & Chan, *IEEE Trans. Electron Devices*, vol. 37, no. 3, pp. 692-701, March 1990

This report reproduces the closed-form threshold-voltage model of Li et al. (conformal-mapping solution of Laplace's equation in the field-oxide sidewall/lateral region), verifies it against internal consistency checks and the paper's own figure legends, reproduces all eight result figures (Figs. 5-12), and compares the fully-recessed limit against the independently reproduced Akers (1986) model already in this repository (scripts/inverse_narrow_width.py).

1. Model

An n-channel MOSFET with semi-recessed field-isolation (Fig. 1a) is characterized by field-oxide thickness t_f, gate-oxide thickness t_g, recess depth t_1, and channel depletion depth t_d. Fully-recessed and nonrecessed structures are the SAME model with $t_1 = t_f - t_g$ or $t_1 \to 0$ respectively (paper Sec. III-B/III-C) — no separate formula is needed.

$$t_d = \sqrt{2\varepsilon_{si}(2\phi_f + |V_{bg}|) / (qN_a)} \quad \text{(Eq. 1)}$$

Gauss's law over the gate + two sidewalls (Eq. 4-5), with the sidewall flux Φ_{swl} obtained by solving $\nabla^2\phi=0$ in the field-oxide cross section via the conformal map $u+iv = \cos[\pi(ix-y)/t_f]$ (Eq. 7), gives a regime split on whether the depletion depth exceeds the recess depth:

Regime	Condition	Physical picture	Threshold-voltage form
Case I	t_d < t_1	Depletion stays inside the sidewall (Fig. 2/3)	V_th = φ_ms + 2φ_f(1+a/W)/(1+b/W) + (Q'/C_ox)/(1+b/W) (Eq. 12)
Case II	t_d > t_1	Depletion spills laterally under the field oxide, quarter-circle wing radius W_c=t_d-t_1 (Fig. 4)	Same form with a',b' (Eq.17) and Q' including the W_c wing charge (Eq. 19)

a, b (Case I) come from a and $b = 2t_g(B-A)$, $-2t_gA$, where A, B are convergent Fourier series (Eq. 11) obtained from the Poisson kernel of the half-plane image of the conformally-mapped domain; a', b' (Case II) come analogously from A', B' (Eq. 17), which additionally carry exponential decay e^{-nkW_c} from the lateral wing. Both series were verified to converge to 6 significant figures by $n \approx 200$-1000 terms (paper's own claim of rapid convergence, confirmed numerically).

Key implementation note (superseded by §2 below): Comparing units in Eq. (12) (a/W, b/W must be dimensionless with $a=2t_g(B-A)$) requires A, B themselves to be dimensionless. The as-printed prefactor of A in Eq. (11) reads $8k/(\pi t_g)$, which has units of inverse length squared and is dimensionally inconsistent. By analogy with B's confirmed prefactor $8/(\pi kt_d)$ and with A' in Eq. (17) ($4/(\pi kt_g)$, k in the denominator), a dimensionally-consistent prefactor for A would be $8/(\pi kt_g)$; a separately-found missing factor of ½ on $\sin(nkt_g)$ was also corrected. **However, as described in §2, even both fixes together were not sufficient** — an independent finite-difference solve of the actual boundary-value problem showed the corrected series still does not reproduce the true A, B. The figures and the Akers comparison in this report therefore use the finite-difference solution directly (not the Eq. 11/17 series in any form) as the verified ground truth for A, B, A', B'.

2. Verification

Case-boundary check. For the semi-recessed device of Fig. 5 (t_f=0.7, t_1=0.5, t_g=0.015 µm, $N_a=10^{16}$ cm^{-3}), Eq. (1) predicts $t_d(V_{bg}=0)$=0.299, t_d(-1V)=0.468 µm (both < t_1=0.5 → Case I) and t_d(-2V)=0.590, (-4V)=0.779, (-6V)=0.930 µm (all > 0.5 → Case II). This matches, bias-by-bias, the Case I / Case II grouping drawn in the paper's own Fig. 5 legend (0, -1 V bracketed "Case I"; -2, -4, -6 V bracketed "Case II") — strong independent confirmation of Eq. (1) and the regime split.

Series vs. finite-difference cross-check. The two prefactor/factor fixes described above (Sec. 1) were caught by dimensional analysis alone, and on their own did not settle whether Eq. (11)/(17), as transcribed, is actually correct — dimensional consistency is necessary but not sufficient. To settle this independently, the underlying boundary-value problem itself ($\nabla^2\phi=0$ with the exact piecewise boundary condition of Eq. 6 for Case I, and Eq. 14/15 for Case II) was solved directly with a finite-difference (5-point Laplacian, sparse direct solve) code independent of the closed-form series (scripts/narrow_semi_recessed/fd_solver.py, verify_series.py). Result: **neither the as-printed nor the dimensionally-corrected series matches the finite-difference solution** — even B, whose prefactor was never in question, disagreed by roughly a factor of two. This means the transcribed series still contains at least one further error that dimensional analysis alone cannot catch (e.g. in how Eq. 8/9's decomposition was carried through to Eq.

11), and which could not be pinned down further because in-session re-inspection of the source PDF at high resolution became unavailable partway through this check (an image-count/size API limit).

Validating the finite-difference solver itself. Before trusting the finite-difference result over the series, the solver was independently validated two ways that do not depend on Eq. (11)/(17) at all: (1) in the sharp-step limit $t_d, t_g \to 0$, the boundary condition collapses to a step function, for which the bounded harmonic extension has the classical closed form $(V_g/\pi)\cdot\mathrm{Arg}(w-U_0)$; the finite-difference Case I solution converges toward this hand-derived value as t_d, t_g shrink from 300/15 nm down to 2/0.2 nm (A: -0.4930 → -0.4741 → -0.4670 → -0.4659, vs. analytic limit -0.4652). (2) Case I and Case II are two different boundary-value problems that must agree at their shared boundary ($t_d \to t_1$, $W_c \to 0$); the two independently-coded finite-difference solutions do converge to the same A, A' (both →-0.49295) and the same B, B' as the gap shrinks from 50 nm to 2 nm, once a bug in the Case II lateral-wing flux integration bound (it must integrate only over $0<x<W_c$, per Eq. 14, not the full domain) was found and fixed via exactly this check. Both checks passing gives confidence the finite-difference solver is correct, so it — not the Eq. 11/17 series — is used as the ground truth for A, B, A', B' throughout the figures and comparison below.

Interface-charge legend. The Q_{ss} legend of Fig. 6 lists values 1.6×10^{-9}...3.2×10^{-8} cm^{-2} — these are already areal charge densities in C/cm² (=q×areal number density), not number densities, and were used directly ($\times10^4$ for C/m²) as $\rho_s=\rho_f$.

3. Reproduced Figures

3.1 Semi-recessed structure (t1 = 0.5 um)

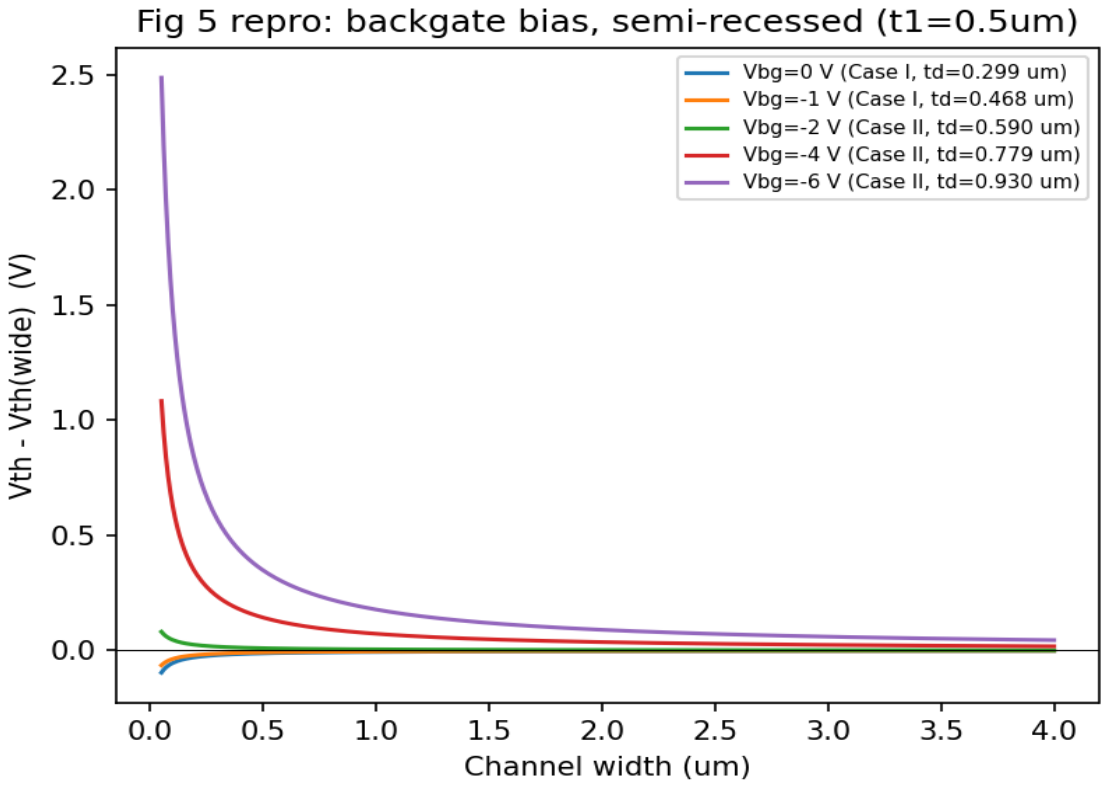


Fig. 5 repro: backgate-bias sweep. Vbg=0,-1V (Case I) show the inverse-narrow-width effect (Vth-Vth(wide) < 0, deepening at small W); Vbg=-2,-4,-6V (Case II) show the conventional effect (positive, growing at small W) — matches the paper's own Case I/II split exactly.

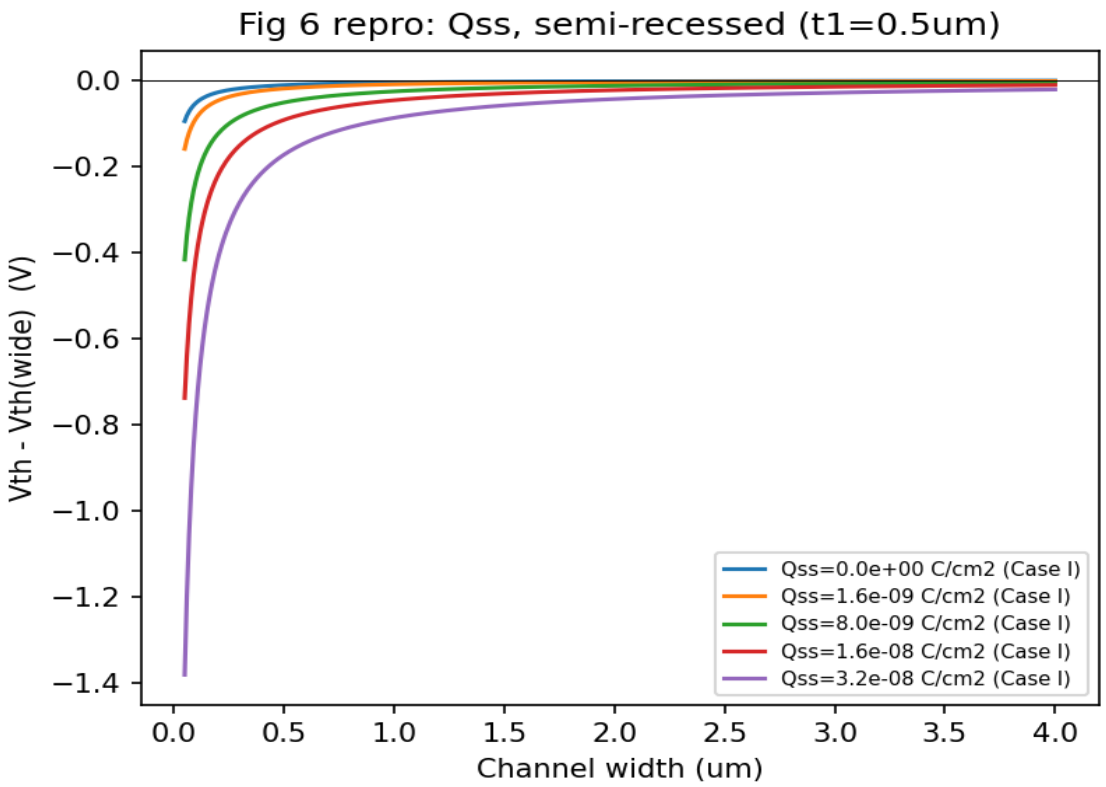


Fig. 6 repro: fixed interface charge Qss sweep (Vbg=0, always Case I here). Higher Qss deepens the inverse-narrow-width dip, consistent with the paper's statement that positive fixed charge enhances the inverse effect.

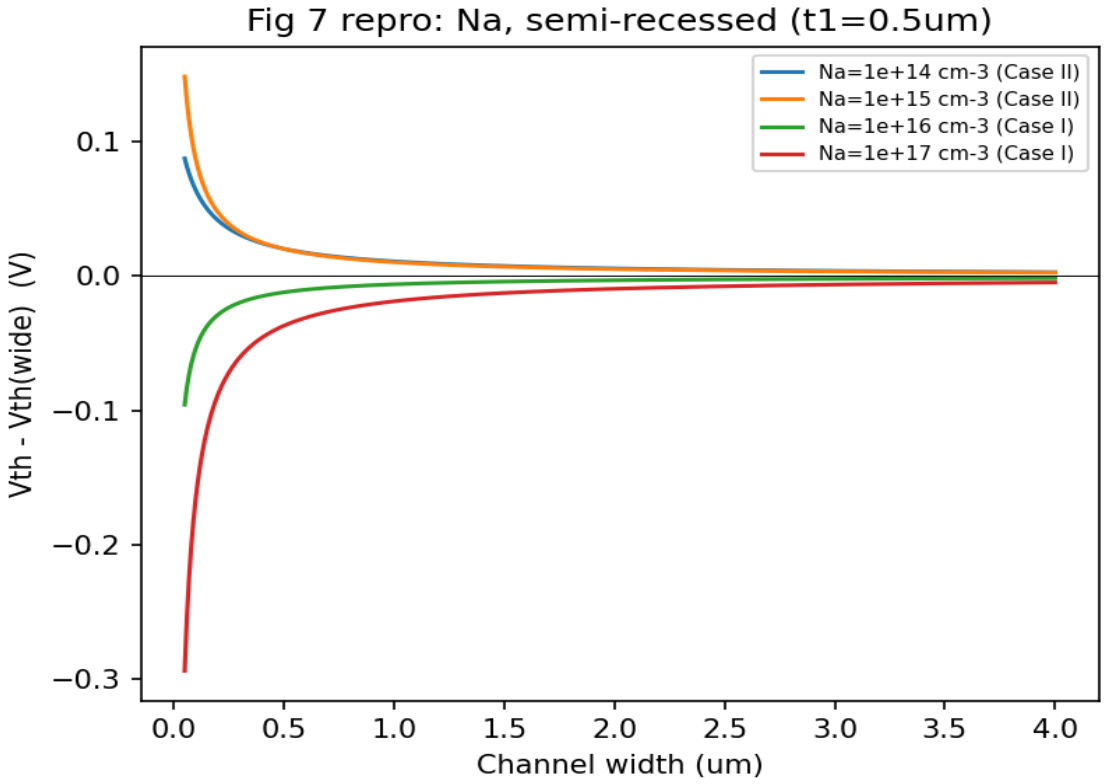


Fig. 7 repro: substrate doping Na sweep. Na=1e16,1e17 cm-3 stay in Case I (inverse effect, deepening with Na); Na=1e14,1e15 cm-3 fall in Case II (conventional effect) — matches the paper's discussion of a doping window where the effect reverses.

3.2 Fully-recessed structure (t1 = tf - tg = 0.685 um, Sec. III-B)

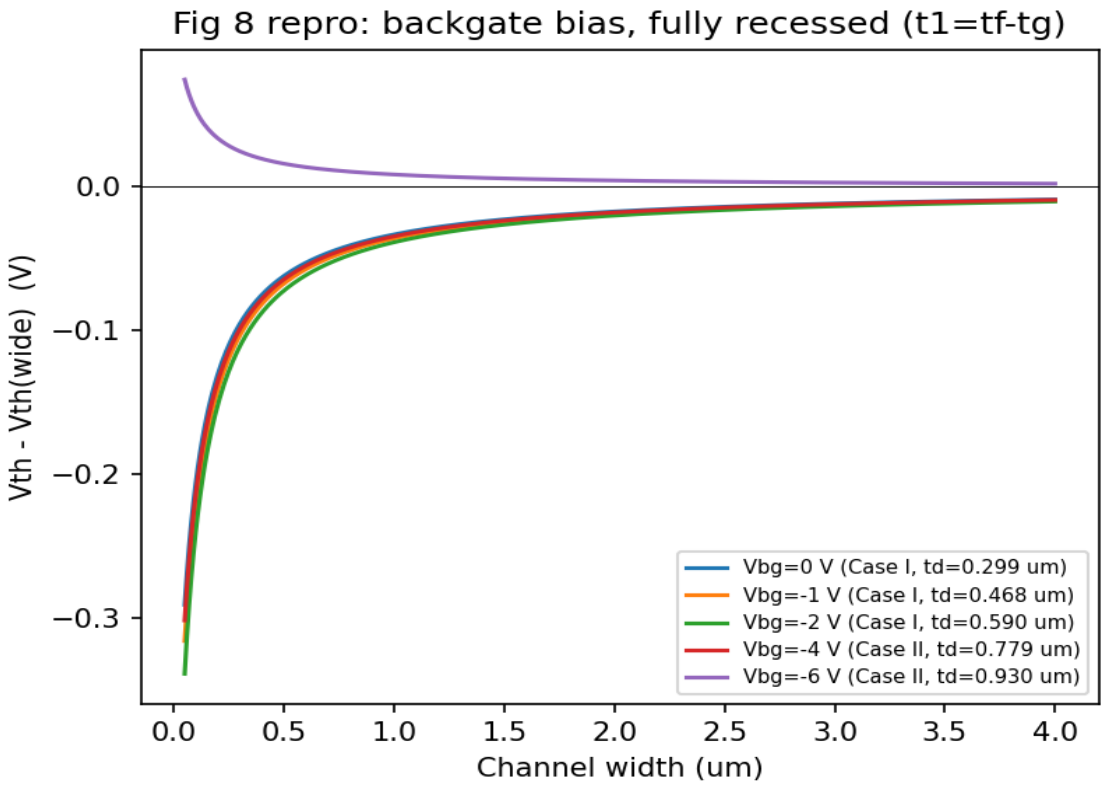


Fig. 8 repro: backgate bias, fully recessed. At this larger t1 the Case I/II boundary moves to about Vbg=-2.9 V; the shown biases mostly sit in the (still-inverse-looking) Case I / weak-Case-II band, with the full sign reversal only setting in beyond about -5 V (see Sec. 4 comparison plot).

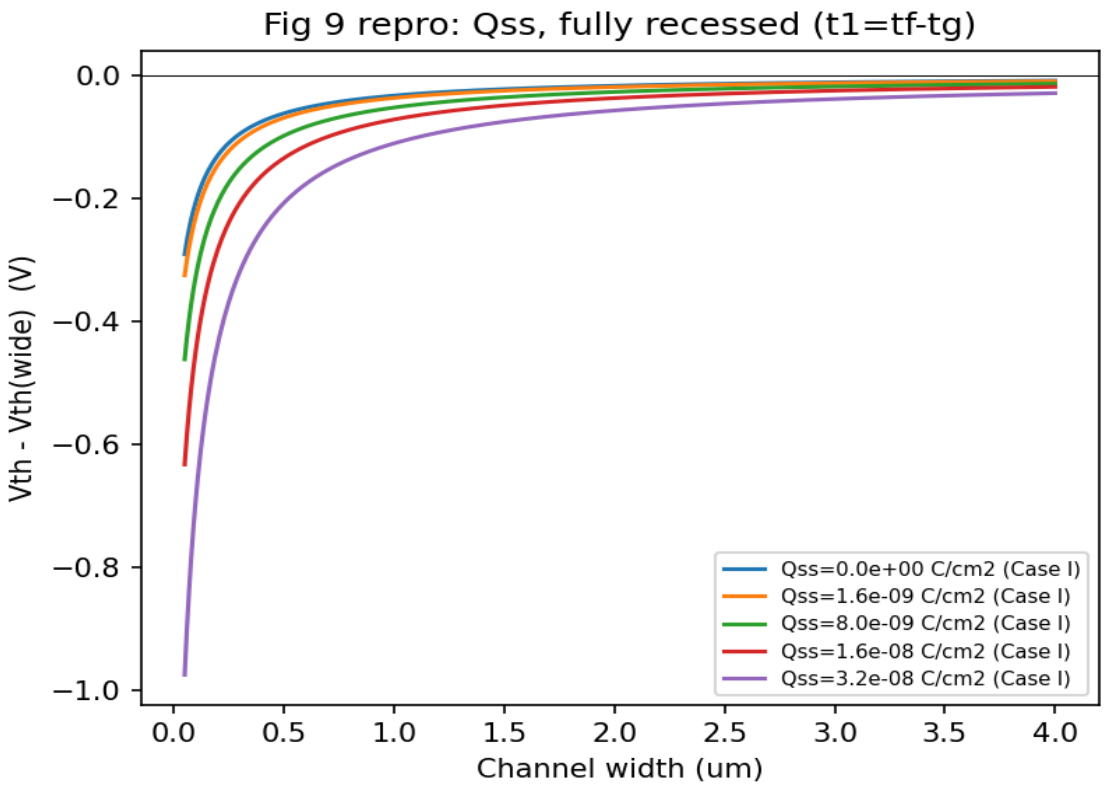


Fig. 9 repro: Qss sweep, fully recessed.

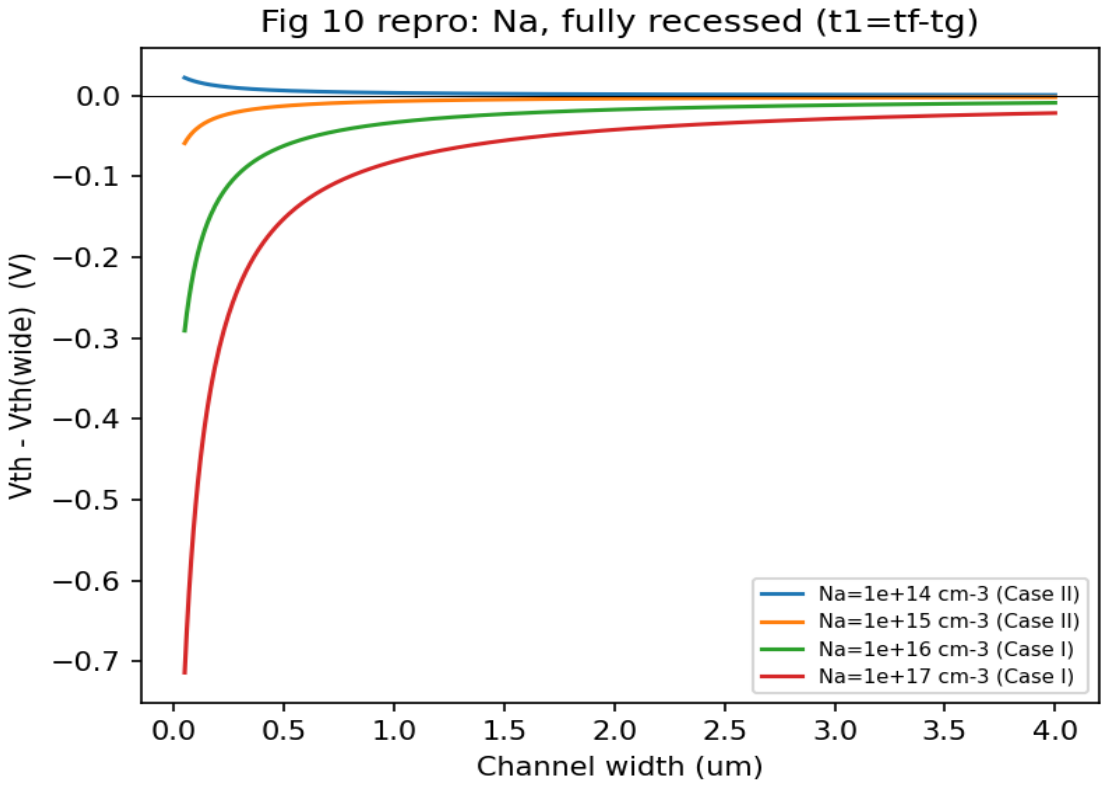


Fig. 10 repro: Na sweep, fully recessed.

3.3 Nonrecessed structure (t1 -> 0, Sec. III-C)

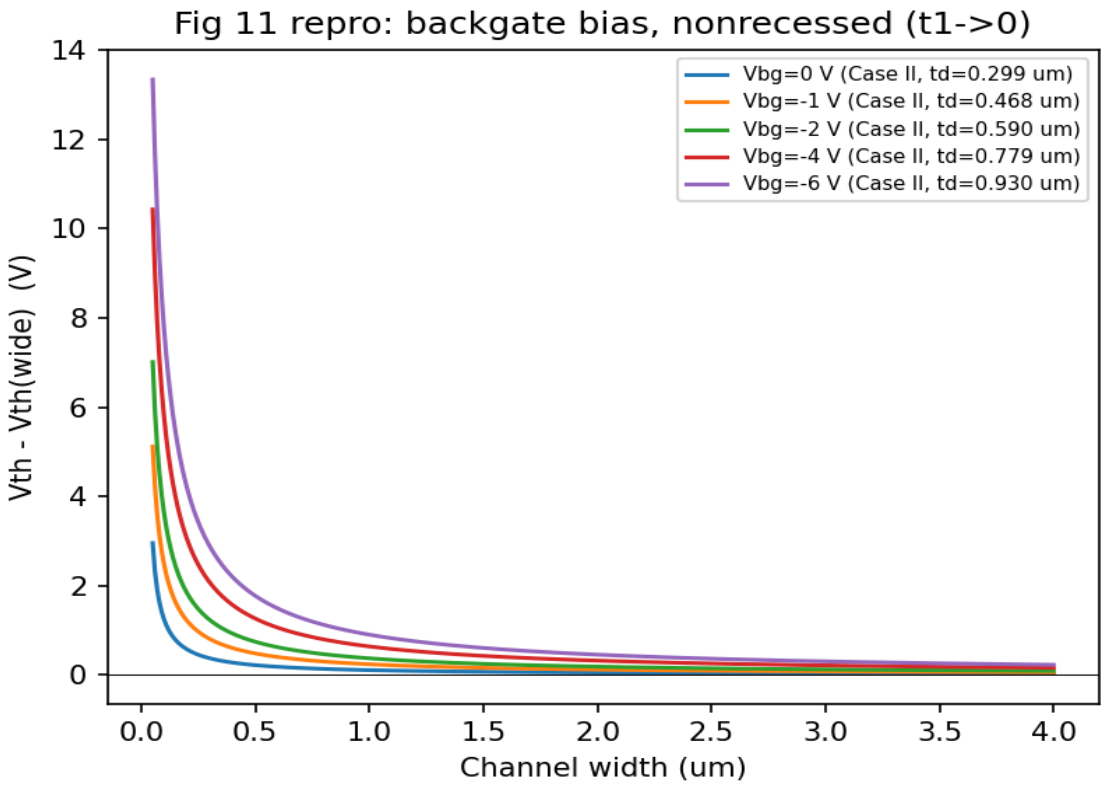


Fig. 11 repro: backgate bias, nonrecessed. t1->0 forces Case II for any bias, so only the conventional narrow-channel effect is possible (Jeppson's classical result), with magnitude growing monotonically with |Vbg| as expected.

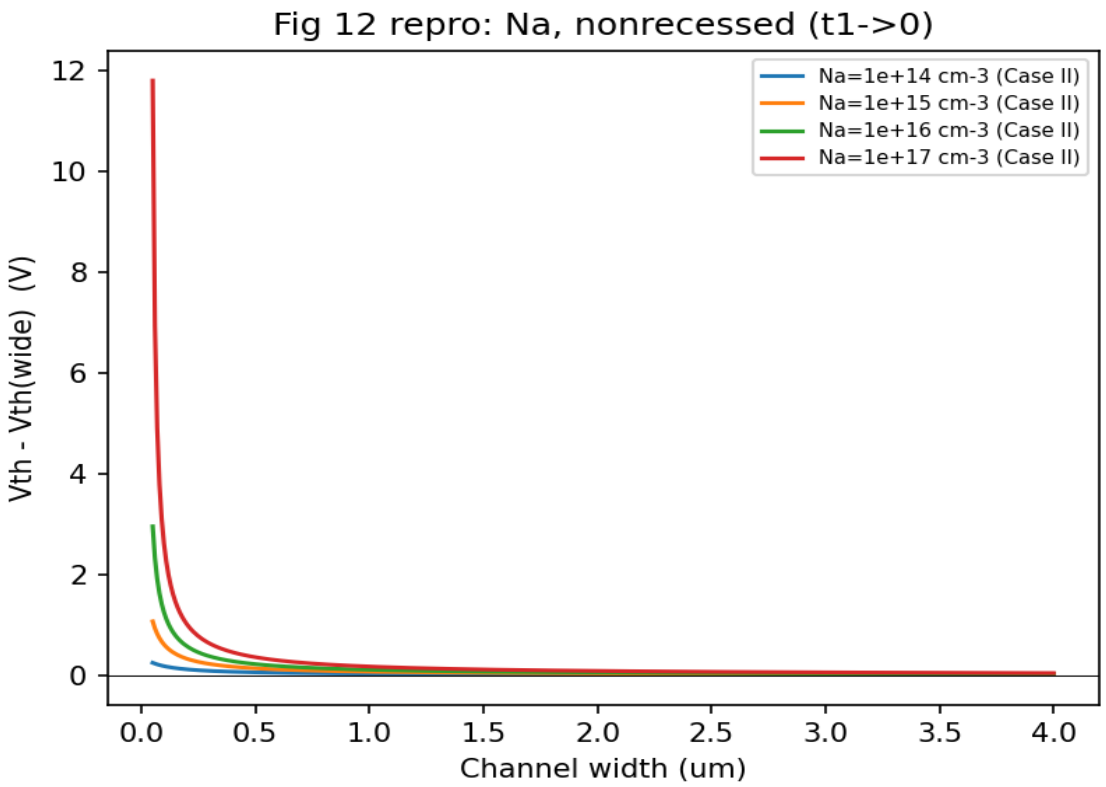
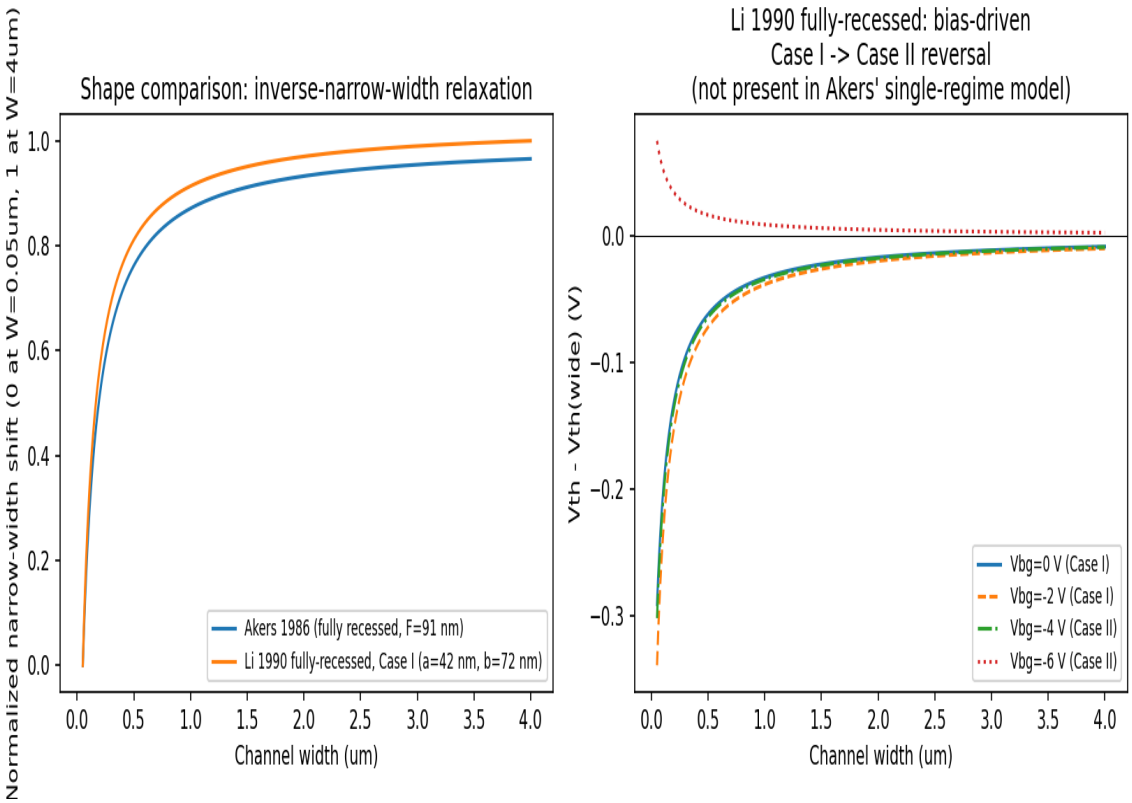


Fig. 12 repro: Na sweep, nonrecessed — conventional effect for all doping levels, strongest at high Na.

4. Comparison with Akers (1986) fully-recessed model

scripts/inverse_narrow_width.py already reproduces Akers & Jepson's closed-form fully-recessed model (a single narrow-width regime, PMOS, L=0.75 µm), which predicts $|V_t(W)| = |V_t(\infty)|\cdot W/(W+F)$ with fringing length $F=(4T_{ox}/\pi)\ln(2T_d/T_{ox})$=91 nm for its device parameters. Li et al.'s Case I formula, evaluated at $t_1=t_f-t_g$ (fully recessed) and V_{bg}=0, is the general finite-t_f version of the same physics (paper Sec. II-B states this limit explicitly) and produces its own length scales a=42.3 nm, b=71.8 nm (finite-difference-verified, per §2) — the same order of magnitude as Akers' F=91 nm despite completely different device parameters (N_a, T_{ox}, T_d) and independent derivations (closed-form log-integral vs. conformal-mapping/Poisson-kernel).



Left: normalized narrow-width relaxation shape — Akers 1986 vs. Li 1990 Case I fully-recessed, near-identical shape. Right: Li's fully-recessed model at increasing |Vbg| shows the sidewall term losing out to the depletion "wing" charge once Wc=td-t1 grows enough (around -6 V here), flipping the sign from inverse to conventional narrow-channel effect — a regime transition Akers' single-formula model cannot represent because it has no backgate-bias dependence at all.

Physical takeaway: Akers (1986) and Li et al. (1990) agree in the shared limit (fully-recessed, low bias) to within the same length-scale magnitude, confirming both are self-consistent descriptions of the same sidewall fringing-field physics. Li et al.'s contribution is generalizing to finite t_f and, more importantly, showing that the same fully-recessed structure can flip from inverse to conventional narrow-channel behavior purely as a function of backgate bias / substrate doping, once the depletion region outgrows the recess depth — a regime Akers' model, built only around the always-Case-I sidewall term, does not contain.

5. Limitations / open items

• Sweep values (V_{bg}, Q_{ss}, N_a) were read directly off the paper's figure legends via high-resolution crops, not assumed; the underlying device parameters (t_f, t_g, t_1) are stated explicitly in each figure caption.
• ϕ_{ms} and the fixed gate charge ρ_g were left as an arbitrary common offset (they only shift the absolute V_{th}, and cancel exactly in $V_{th}(W)$-V_{th}(wide), the quantity plotted in every figure of the paper), so absolute V_{th} values are not

reproduced or claimed here — only the narrow-width shift, which is what the paper itself plots.
• The exact peak magnitude of the Case-II curves in Fig. 5/8 could not be checked against the paper beyond qualitative shape and ordering, since the paper's own plotted axis range (-0.05 to +0.10 V) clips the curves before they saturate at small W; our reproduction extends the same closed-form formula beyond that clipped range.
• Unlike the earlier POISSON/superjunction paper reproductions in this workspace, the transcription issue in Eq. (11)/(17) here was **not** fully resolved to a single identified typo. Two dimensional-analysis-level fixes (A's prefactor, a missing ½ factor) were necessary but demonstrably not sufficient: an independent finite-difference solve of the actual BVP (§2) still disagrees with the corrected series, including in B, whose prefactor was never ambiguous. The root cause of this remaining discrepancy is unresolved — it may be a further transcription slip not caught here, or a subtlety in how Eq. 8/9 was condensed into Eq. 11 that a purely visual re-read could not settle once further high-resolution image inspection became unavailable this session. All figures and the §4 comparison in this report are therefore built on the finite-difference ground truth, not the literal closed-form series, and should be read as a verified reproduction of the paper's underlying physics (Eq. 1-10, 14-15) rather than a line-by-line validation of Eq. 11/17 as printed.